MILITARY SPECIFICATION SHEET

INSULATION SLEEVING, ELECTRICAL, FLEXIBLE, COATED

CLASS 220, TYPE E, CATEGORY b

The complete requirements for procuring the insulation sleeving described herein shall consist of this document and the latest issue of Specification MIL-I-3190.

The sleeving shall be of the following classification:

Class 220, Type E, Category b

REQUIREMENTS:

The insulation sleeving shall be in accordance with tables I, II, and III.

Table I - Performance requirements.

Property	Conditioning prior to test	Test method paragraph	Unit	Value required
Dielectric breakdown:				
Straight	96/23/50	4.7.2.2	Volts(minimum average)	7000
Straight	96/23/50	4.7.2.2	Volts(minimum individual)	5000
Straight	96/23/96	4.7.2.2	Percent of average value obtained on test after condition 96/23/50 (minimum)	50
Straight	At temp. of 220°C	4.7.2.2	Volts (minimum average)	5000
90° bend	96/23/50	4.7.2.3	Volts (minimum average)	5500
Flammability: Burning Time	96/23/50	4.7.5.1 Method A	Seconds (maximum average)	10
Length burned			(see 4.6.5)	The paper indicator shall not be affected
Oil and solvent resistance: Swelling oil	96/23/50	4.7.6.2	Observation	Shall not disintegrate, peel or crack
Xylol	96/23/50	4.7.6.2	Observation	Shall not disintegrate, peel or crack
1,1,1 Trichloroethane (inhibited)	96/23/50	4.7.6.2	Observation	Shall not disintegrate, peel or crack

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Table I - Performance requirements. (Cont'd.)

Property	Conditioning prior to test	Test method paragraph	Unit	Value required
Thermal endurance	see 4.7.9	4.7.9	Temp. index(°C.) at 15000 hours (minimum)	220
Thermal stability	96/275	4.7.10	Volts (minimum average)	5000

Table II - Qualification inspection requirements.

Examination or test	Requirement reference	Test reference	Minimum number of tests for each size and for each condition on table I
Visual examination	3.3, 3.6	4.5	10
Dimensions	3.4	4.7	10
Dielectric breakdown:			
Straight - 96/23/50, 96/23/96 and at temperature	table I	4.7.2.2	10
90° bend	table I	4.7.2.3	5
Flammability	table I	4.7.5.1 Method A	5
Oil and solvent resistance: Swelling oil	table I	4.7.6.2	3
Xylol	table I	4.7.6.2	3
1,1,1 - Trichlorethane (inhibited)	table I	4.7.6.2	3
Thermal endurance	table I	4.7.9	see 4.7.9

Table III - Quality conformance inspection requirements.

Examination or test	Requirement reference	Test reference	Conditioning prior to test	Minimum number of tests per sample unit
Visual examination	3.3, 3.6	4.5	96/23/50	As required per sampling
Dimensions	3.4	4.7	96/23/50	As required per sampling
Dielectric breakdown: Straight	table I	4.7.2.2	96/23/50 96/23/96	10 10
Flammability	table I	4.7.5.1 Method A	96/23/50	5
Solvent resistance: Xylol	table I	4.7.6.2	96 hours immersion at 23°C	3
Thermal stability	table I	4.7.10	96/275	5

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Notes

Intended use. The temperature index for this class of sleeving is 220. This material is intended for applications in electrical equipment requiring primary insulation having high thermal capability, flame and solvent resistance, and providing adequate mechanical protection. It may be used where additional protection against environmental conditions or protection in localized areas is desired. It is to be noted that the polyimide sleeving is considerably stiffer and of a lower order of flexibility than other sleeving.

Preparing activity:
Navy - SH
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